



Confirm data-cleaning meets business expectations

In this video, we'll discuss how to begin the process of verifying your data-cleaning efforts. Verification is a critical part of any analysis project. Without it you have no way of knowing that your insights can be relied on for data-driven decision-making. Think of verification as a stamp of approval. To refresh your memory, **verification is a process to confirm that a data-cleaning effort was well-executed and the resulting data is accurate and reliable.** It also involves manually cleaning data to compare your expectations with what's actually present. The first step in the verification process is going back to your original unclean data set and comparing it to what you have now.

Review the dirty data and try to identify any common problems. For example, maybe you had a lot of nulls. In that case, you check your clean data to ensure no nulls are present. To do that, you could search through the data manually or use tools like conditional formatting or filters. Or maybe there was a common misspelling like someone keying in the name of a product incorrectly over and over again. In that case, you'd run a FIND in your clean data to make sure no instances of the misspelled word occur. Another key part of verification involves taking a **big-picture view** of your project. This is an opportunity to confirm you're actually focusing on the business problem that you need to solve and the overall project goals and to make sure that your data is actually capable of solving that problem and achieving those goals. It's important to take

the time to reset and focus on the big picture because projects can sometimes evolve or transform over time without us even realizing it. Maybe an e-commerce company decides to survey 1000 customers to get information that would be used to improve a product. But as responses begin coming in, the analysts notice a lot of comments about how unhappy customers are with the e-commerce website platform altogether. So the analysts start to focus on that. While the customer buying experience is of course important for any e-commerce business, it wasn't the original objective of the project. The analysts in this case need to take a moment to pause, refocus, and get back to solving the original problem.

Taking a big picture view of your project involves doing three things. First, consider the business problem you're trying to solve with the data. If you've lost sight of the problem, you have no way of knowing what data belongs in your analysis. Taking a problem-first approach to analytics is essential at all stages of any project. You need to be certain that your data will actually make it possible to solve your business problem. Second, you need to consider the goal of the project. It's not enough just to know that your company wants to analyze customer feedback about a product.

What you really need to know is that the goal of getting this feedback is to make improvements to that product. On top of that, you also need to know whether the data you've collected and cleaned will actually help your company achieve that goal. And third, you need to consider whether your data is capable of solving the problem and meeting the project objectives. That means thinking about where the data came from and testing your data collection and cleaning processes. Sometimes data analysts can be too familiar with their own data, which makes it easier to miss something or make

assumptions. Asking a teammate to review your data from a fresh perspective and getting feedback from others is very valuable in this stage. This is also the time to notice if anything sticks out to you as suspicious or potentially problematic in your data. Again, step back, take a big picture view, and ask yourself, do the numbers make sense? Let's go back to our e-commerce company example. Imagine an analyst is reviewing the cleaned up data from the customer satisfaction survey. The survey was originally sent to 1,000 customers, but what if the analyst discovers that there is more than a thousand responses in the data? This could mean that one customer figured out a way to take the survey more than once. Or it could also mean that something went wrong in the data cleaning process, and a field was duplicated. Either way, this is a signal that it's time to go back to the data-cleaning process and correct the problem. Verifying your data ensures that the insights you gain from analysis can be trusted. It's an essential part of data-cleaning that helps companies avoid big mistakes. This is another place where data analysts can save the day. Coming up, we'll go through the next steps in the data-cleaning process. See you there.